

Muhammad Shaheer | FA24-BSE-104(B)

Object Oriented Programming

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Object-Oriented Programming Concepts using Classes, Objects, and Constructors in Java

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ASSIGNMENT#01

Q.NO.1. WRITE A CLASS STUDENT WITH DATA MEMBERS STUDENT NAME, ROLL NO., SEMESTER NO., COURSE NAME, MARKS, GRADE AND MAKE THEM AS PRIVATE. WRITE MEMBER FUNCTIONS WITH TWO CONSTRUCTORS ONE MUST BE PARAMETERIZED THAT ASSIGNS VALUES TO THE DATA MEMBERS OF THE CLASS. ONE MEMBER FUNCTION GET_STUDENT_DATA() SHOULD GET THE STUDENT DATA FROM THE USER. DISPLAY_QUIZ_GRADE() FUNCTION SHOULD DISPLAY THE STUDENT'S GRADE ACCORDING TO MARKS. YOU HAVE TO TAKE THREE STUDENTS' DATA ONE FROM USER AND TWO FROM CONSTRUCTORS THAT INITIALIZES STUDENTS' DATA.

```
import java.util.*;

class Student {
    private String studentName, rollNo, courseName;
    private int semesterNo;
    private double marks;
    private char grade;

    public Student() {
        studentName = "Null";
        rollNo = "Null";
        courseName = "Not Registered";
        semesterNo = 0;
        marks = 0.0;
        grade = 'N';
    }

    public Student(String studentName, String rollNo, String
courseName, int semesterNo, double marks) {
        this.studentName = studentName;
        this.rollNo = rollNo;
        this.courseName = courseName;
        this.semesterNo = semesterNo;
        this.marks = marks;
        this.grade = calculateGrade(marks);
    }

    void get_student_data() {
        Scanner in = new Scanner(System.in);
        System.out.print("Enter student name = ");
        studentName = in.nextLine();
        System.out.print("Enter roll no. = ");
        rollNo = in.nextLine();
        System.out.print("Enter semester no. = ");
```

```

        semesterNo = in.nextInt();
        in.nextLine();
        System.out.print("Enter course name = ");
        courseName = in.nextLine();
        System.out.print("Your marks is = ");
        marks = in.nextDouble();
        grade = calculateGrade(marks);
    }

    private char calculateGrade(double marks) {
        if (marks >= 9)
            return 'A';
        else if (marks >= 7.5)
            return 'B';
        else if (marks >= 6.5)
            return 'C';
        else if (marks >= 5.5)
            return 'D';
        else if (marks >= 4.5)
            return 'E';
        else
            return 'F';
    }

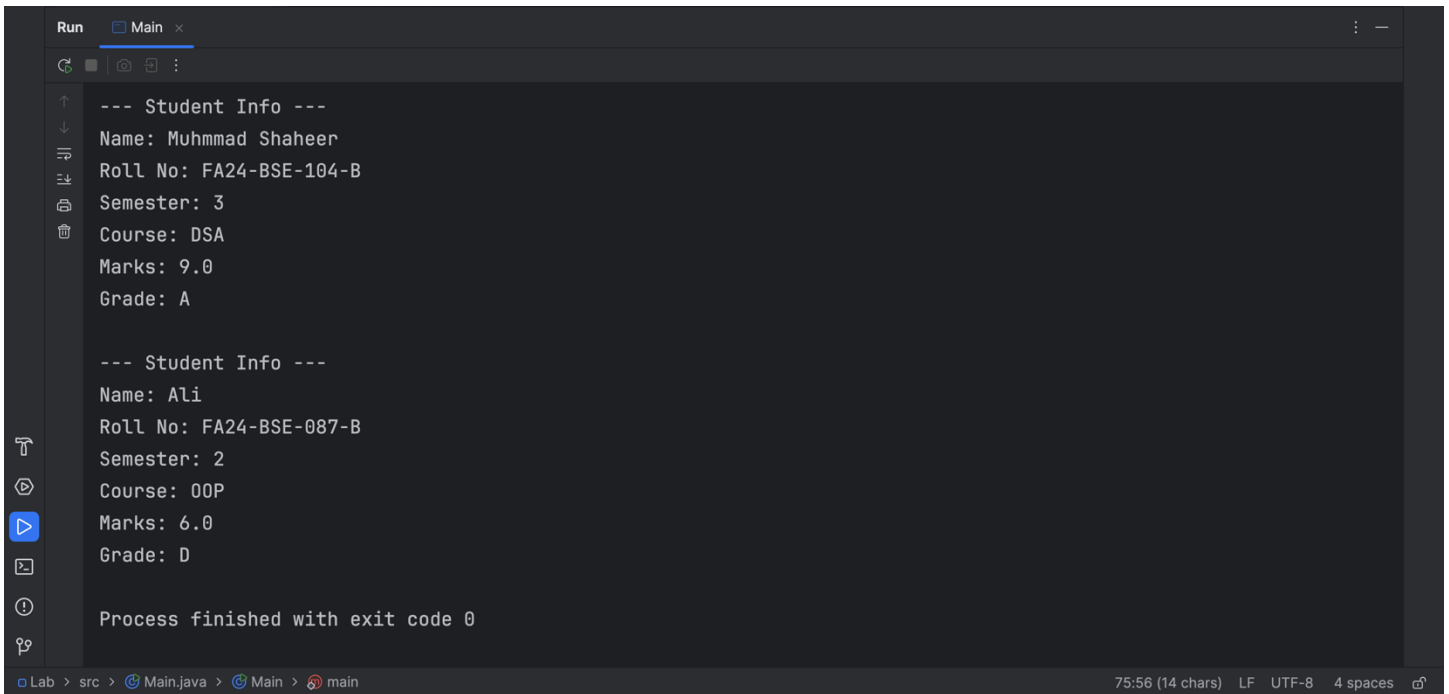
    void Display_quiz_grade() {
        System.out.println("\n--- Student Info ---");
        System.out.println("Name: " + studentName);
        System.out.println("Roll No: " + rollNo);
        System.out.println("Semester: " + semesterNo);
        System.out.println("Course: " + courseName);
        System.out.println("Marks: " + marks);
        System.out.println("Grade: " + grade);
    }
}

public class Main{
    public static void main(String[] args) {
        System.out.println("Enter student1 data");
        Student s1 = new Student();
        s1.get_student_data();

        Student s2 = new Student("Ali", "FA24-BSE-087-B",
"OOP", 2, 6);
    }
}

```

```
s1.Display_quiz_grade();  
s2.Display_quiz_grade();  
}  
}
```



```
Run Main x  
--- Student Info ---  
Name: Muhmmad Shaheer  
Roll No: FA24-BSE-104-B  
Semester: 3  
Course: DSA  
Marks: 9.0  
Grade: A  
  
--- Student Info ---  
Name: Ali  
Roll No: FA24-BSE-087-B  
Semester: 2  
Course: OOP  
Marks: 6.0  
Grade: D  
  
Process finished with exit code 0  
Lab > src > Main.java > Main > main 75:56 (14 chars) LF UTF-8 4 spaces
```

Q2. WRITE A CLASS TRAVEL THAT HAS ATTRIBUTES OF KILOMETERS AND HOURS. A CONSTRUCTOR WITH NO PARAMETERS INITIALIZES BOTH DATA MEMBERS TO 0.

A MEMBER FUNCTION GET() INPUTS THE VALUES AND FUNCTION SHOW() DISPLAY THE VALUES.

IT HAS A MEMBER FUNCTION ADD() THAT TAKES AN OBJECT OF TYPE TRAVEL TO ADD THE KILOMETERS AND HOURS OF CALLING OBJECT AND PARAMETER.

```
import java.util.Scanner;

class Travel {
    private int kilometers;
    private int hours;

    public Travel() {
        kilometers = 0;
        hours = 0;
    }

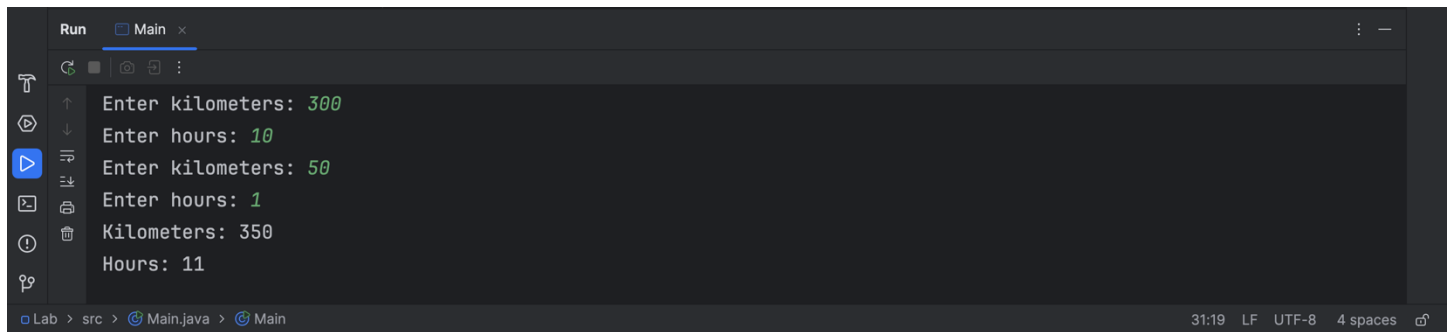
    void get() {
        Scanner input = new Scanner(System.in);
        System.out.print("Enter kilometers: ");
        kilometers = input.nextInt();
        System.out.print("Enter hours: ");
        hours = input.nextInt();
    }

    void show() {
        System.out.println("Kilometers: " + kilometers);
        System.out.println("Hours: " + hours);
    }

    void add(Travel t) {
        this.kilometers += t.kilometers;
        this.hours += t.hours;
    }
}

public class Main{
    public static void main(String[] args) {
        Travel t1 = new Travel();
        Travel t2 = new Travel();
        t1.get();
        t2.get();
    }
}
```

```
        t1.add(t2);  
        t1.show();  
    }  
}
```



The screenshot shows an IDE's Run console window. The title bar reads "Run" and "Main x". On the left is a vertical toolbar with icons for Run, Step Over, Step Into, Step Out, Breakpoints, and Debug Console. The main area displays the following text: "Enter kilometers: 300", "Enter hours: 10", "Enter kilometers: 50", "Enter hours: 1", "Kilometers: 350", and "Hours: 11". The bottom status bar shows the file path "Lab > src > Main.java > Main", the time "31:19", and encoding/formatting details "LF UTF-8 4 spaces".

```
Run Main x  
Enter kilometers: 300  
Enter hours: 10  
Enter kilometers: 50  
Enter hours: 1  
Kilometers: 350  
Hours: 11  
Lab > src > Main.java > Main 31:19 LF UTF-8 4 spaces
```

Q3. WRITE A CLASS TRAVEL THAT HAS ATTRIBUTES OF KILOMETERS AND HOURS. A CONSTRUCTOR WITH NO PARAMETERS INITIALIZES BOTH DATA MEMBERS TO 0.

A MEMBER FUNCTION GET() INPUTS THE VALUES AND FUNCTION SHOW() DISPLAY THE VALUES.

IT HAS A MEMBER FUNCTION ADD() THAT TAKES AN OBJECT OF TYPE TRAVEL TO ADD THE KILOMETRES AND HOURS OF CALLING OBJECT AND PARAMETER AND RETURNS AN OBJECT WITH THE ADDED VALUE.

```
import java.util.*;

class Travel {
    private double kilometers;
    private int hours;

    public Travel() {
        this.kilometers = 0;
        this.hours = 0;
    }

    void get(){
        Scanner in = new Scanner(System.in);
        System.out.print("Enter kilometers: ");
        kilometers = in.nextDouble();
        System.out.print("Enter hours: ");
        hours = in.nextInt();
    }

    public void show(){
        System.out.println("Kilometers: " + kilometers);
        System.out.println("Hours: " + hours);
    }

    public Travel add(Travel t){
        Travel result = new Travel();
        result.kilometers = this.kilometers + t.kilometers;
        result.hours = this.hours + t.hours;
        return result;
    }
}

public class Main{
    public static void main(String[] args){
        Travel t1 = new Travel();
        Travel t2 = new Travel();
    }
}
```

```

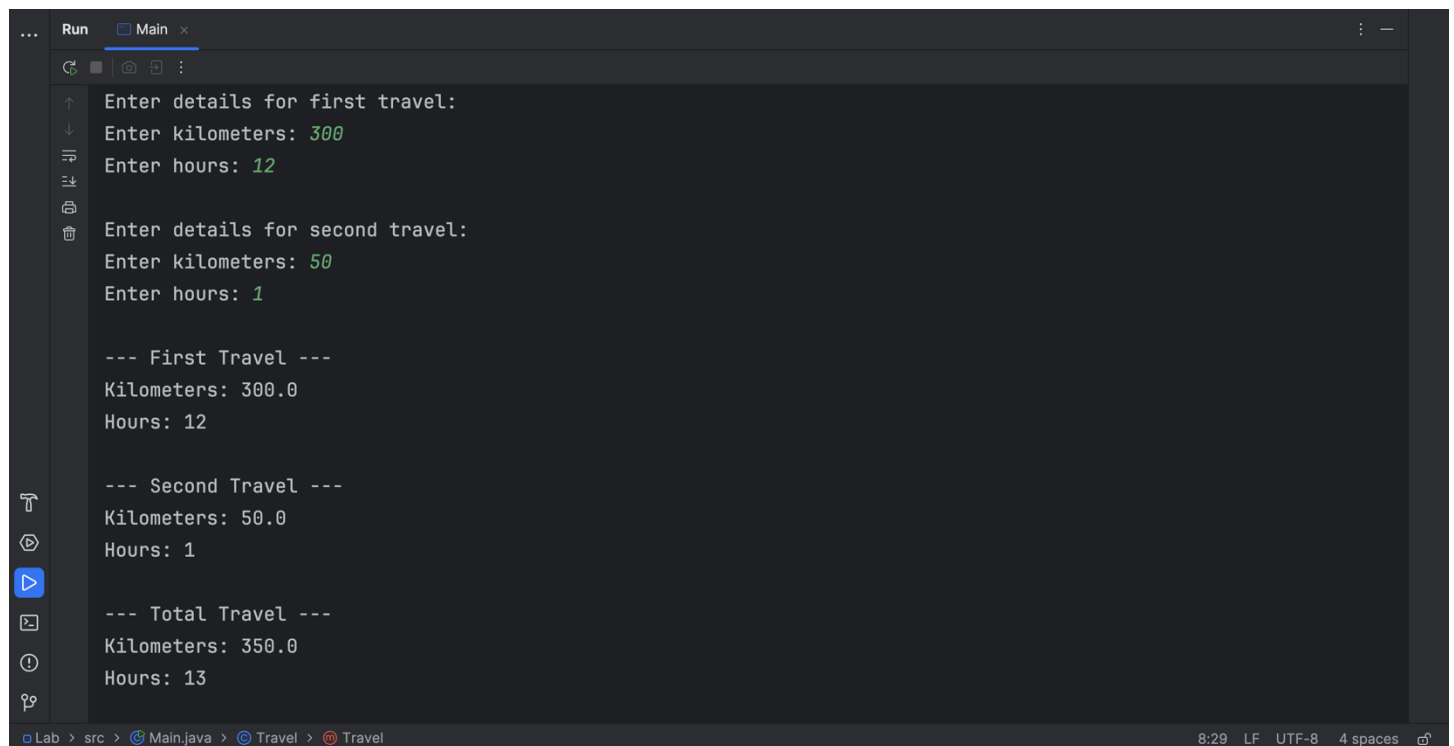
        System.out.println("Enter details for first travel:");
        t1.get();
        System.out.println("\nEnter details for second
travel:");
        t2.get();

        Travel total = t1.add(t2);

        System.out.println("\n--- First Travel ---");
        t1.show();
        System.out.println("\n--- Second Travel ---");
        t2.show();

        System.out.println("\n--- Total Travel ---");
        total.show();
    }
}

```



The screenshot shows an IDE with a 'Run' window open. The output of the Java program is as follows:

```

Enter details for first travel:
Enter kilometers: 300
Enter hours: 12

Enter details for second travel:
Enter kilometers: 50
Enter hours: 1

--- First Travel ---
Kilometers: 300.0
Hours: 12

--- Second Travel ---
Kilometers: 50.0
Hours: 1

--- Total Travel ---
Kilometers: 350.0
Hours: 13

```

The IDE interface includes a sidebar with icons for Explorer, Run and Debug, Source, and Run and Debug. The bottom status bar shows the file path 'Lab > src > Main.java > Travel > Travel', the time '8:29', and encoding 'UTF-8'.